Experiment No. 2

Signal Operations and Transformations

2.2. To perform time shifting and time reversal of continuous-time and discrete-time signals and analyze the resulting transformations

Experimental Procedure

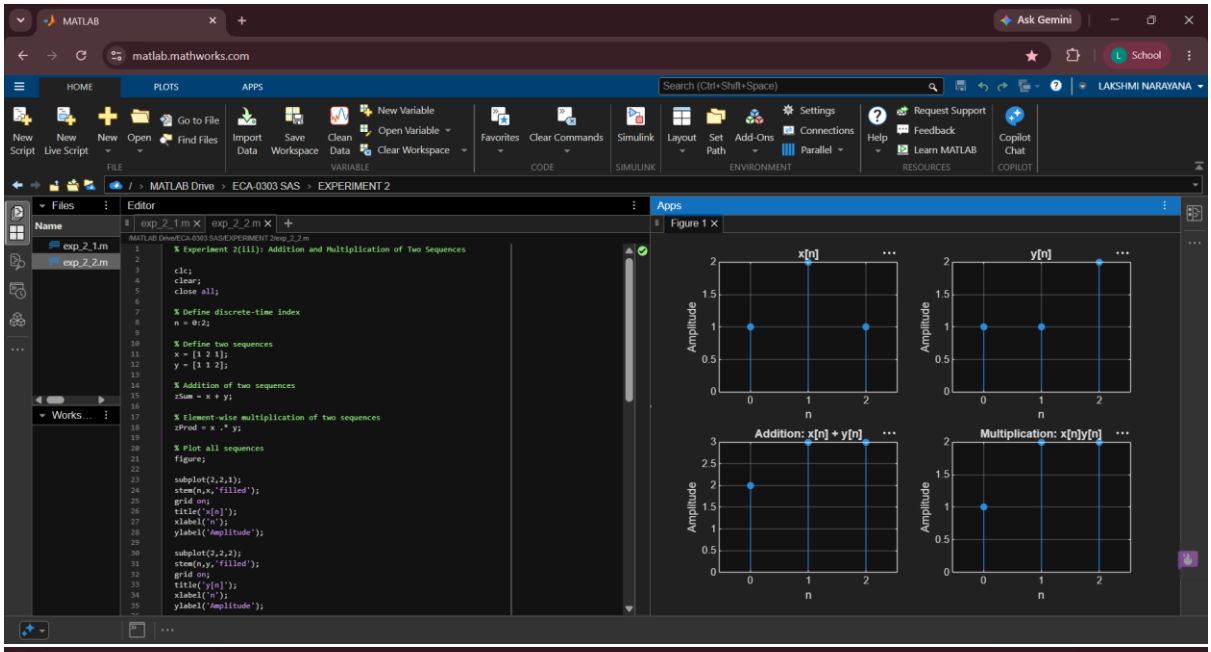
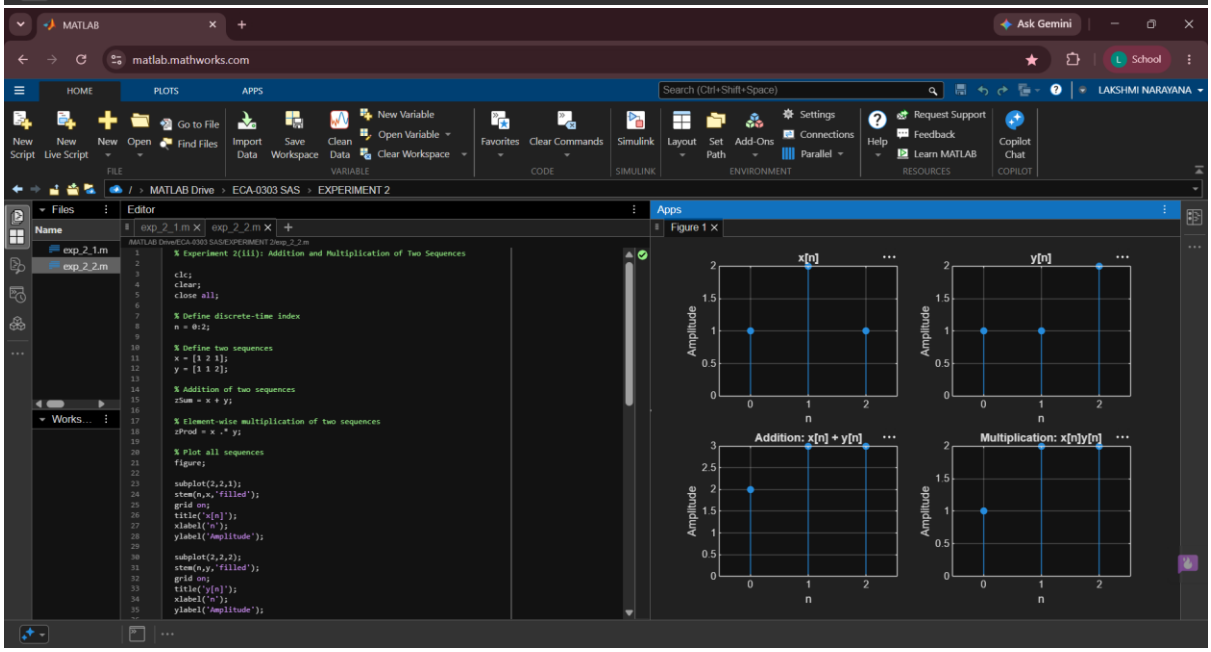
1. Generate the base signal x(t).
2. Create the delayed signal $x(t-t_0)$.
3. Create the time-reversed signal $x(-t)$.
4. Plot all signals using common axes.
5. Measure or identify the time displacement.
6. Compare the transformed waveforms with theoretical expectations.

MATLAB Coding

```
% Experiment 2(ii): Time Shifting and Time Reversal
clc; clear; close all;
t = -1:0.001:1; t0 = 0.2;
x = sin(2*pi*5*t);
xShift = sin(2*pi*5*(t-t0));
xRev = sin(2*pi*5*(-t));

figure;
subplot(3,1,1); plot(t,x); grid on;
title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(3,1,2); plot(t,xShift); grid on;
title('Time Shifted x(t-t0)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(3,1,3); plot(t,xRev); grid on;
title('Time Reversed x(-t)'); xlabel('Time (s)'); ylabel('Amplitude');
```

Output:


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exp_2_2.m
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% Experiment 2(iii): Addition and Multiplication of Two Sequences
clc;
clear;
close all;
% Define discrete-time index
n = 0:2;
% Define two sequences
x = [1 2 1];
y = [1 1 2];
% Addition of two sequences
zSum = x + y;
% Element-wise multiplication of two sequences
zProd = x .* y;
% Plot all sequences
figure;
subplot(2,2,1);
stem(n,x,'filled');
grid on;
title('x[n]');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,2);
stem(n,y,'filled');
grid on;
title('y[n]');
xlabel('n');
ylabel('Amplitude');
Apps
Figure 1
x[n]
y[n]
Addition: x[n] + y[n]
Multiplication: x[n]y[n]
Amplitude
n

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% Plot all sequences
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subplot(2,2,1);
stem(n,x,'filled');
grid on;
title('x[n]');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,2);
stem(n,y,'filled');
grid on;
title('y[n]');
xlabel('n');
ylabel('Amplitude');
Apps
Figure 1
x[n]
y[n]
Addition: x[n] + y[n]
Multiplication: x[n]y[n]
Amplitude
n